Chapter 3. Inference for Contingency Tables

Chi-squared tests, ordinal trends, exact tests

Contents

3.1 Confidence intervals for association parameters **2**
- 3.1.1 Interval estimation of odds ratios . . . 2
- 3.1.2 Aspirin and myocardial infarction example . . . 2
- 3.1.3 Interval estimation of difference of proportions . . . 2
- 3.1.4 Interval estimation of relative risk . . . 3
- 3.1.5 Deriving standard errors with the delta method* . . . 3
- 3.1.6 Delta method applied to sample logit* . . . 4
- 3.1.7 Delta method for log odds ratio* . . . 4

3.2 Testing independence in two-way contingency tables **5**
- 3.2.1 Pearson and likelihood-ratio chi-squared tests . . . 5
- 3.2.2 Education and religious fundamentalism example . . . 6

3.3 Following-up chi-squared tests **6**
- 3.3.1 Pearson and standardized residuals . . . 6
- 3.3.2 Education and religious fundamentalism revisited . . . 7
- 3.3.3 Partitioning chi-squared . . . 7
- 3.3.4 Origin of schizophrenia example . . . 8
- 3.3.5 Rules for partitioning . . . 8
- 3.3.6 Limitations of chi-squared tests . . . 8
- 3.3.7 Why consider independence? . . . 8

3.4 Two-way tables with ordered classifications **9**
- 3.4.1 Linear trend alternative to independence . . . 9
- 3.4.2 Job satisfaction example revisited . . . 9
- 3.4.3 Monotone trend alternatives to independence . . . 9
- 3.4.4 Extra power with ordinal tests . . . 10
- 3.4.5 Choice of scores . . . 10
- 3.4.6 Trend tests for $I \times 2$ and $2 \times J$ tables . . . 10
- 3.4.7 Nominal–ordinal tables . . . 10

3.5 Small-sample tests of independence **10**
- 3.5.1 Fisher's exact test for 2×2 tables . . . 10
- 3.5.2 Fisher's tea drinker . . . 11
- 3.5.3 Two-sided P-values for Fisher's exact test . . . 11
- 3.5.4 Discreteness and conservatism issues . . . 12

3.6 Extensions for multiway tables and nontabulated responses **12**
- 3.6.1 Categorical data need not be contingency tables . . . 12

3.1 Confidence intervals for association parameters

Motivation. Standard errors for association measures come from the delta method; better intervals come from inverting score or LR tests.

3.1.1 Interval estimation of odds ratios

The sample odds ratio $\hat{\theta} = n_{11}n_{22}/(n_{12}n_{21})$ is 0 or ∞ when some $n_{ij} = 0$, so $E(\hat{\theta})$ and $\text{Var}(\log\hat{\theta})$ do not exist; the amended $\tilde{\theta} = \frac{(n_{11}+0.5)(n_{22}+0.5)}{(n_{12}+0.5)(n_{21}+0.5)}$ has the same asymptotic distribution and behaves better. $\hat{\theta}$ is very skewed, while $\log\hat{\theta}$ approaches normality faster, with estimated standard error (Corollary 3.7)

$$\hat{\sigma}(\log\hat{\theta}) = \Big(\frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}\Big)^{1/2}. \tag{3.1}$$

Definition (Woolf interval)

$\log\hat{\theta} \pm z_{\alpha/2}\,\hat{\sigma}(\log\hat{\theta})$ is a Wald interval for $\log\theta$; exponentiating its endpoints gives the Woolf interval for θ.

It is usually slightly conservative. If $\hat{\theta} = 0$ (or ∞), use 0 (or ∞) as one limit and compute the other with $n_{ij} + 0.5$.

3.1.2 Aspirin and myocardial infarction example

Example 3.1 (Swedish aspirin study: odds ratio)

Of 684 placebo patients 28 died of MI; of 676 aspirin patients, 18. Then $\hat{\theta} = \frac{28\times658}{656\times18} = 1.56$, $\log\hat{\theta} = 0.445$, $\hat{\sigma}(\log\hat{\theta}) = 0.307$, and the 95% interval for θ is $(e^{-0.157}, e^{1.047}) = (0.85, 2.85)$: equal odds are plausible.

3.1.3 Interval estimation of difference of proportions

Treat the rows as independent binomials $y_i \sim \text{bin}(n_i, \pi_i)$, $\hat{\pi}_i = y_i/n_i$.

Proposition 3.2 (Difference of proportions)

$E(\hat{\pi}_1 - \hat{\pi}_2) = \pi_1 - \pi_2$ and

$$\sigma(\hat{\pi}_1 - \hat{\pi}_2) = \Big[\frac{\pi_1(1-\pi_1)}{n_1} + \frac{\pi_2(1-\pi_2)}{n_2}\Big]^{1/2}. \tag{3.2}$$

As $n_1, n_2 \to \infty$, $(\hat{\pi}_1 - \hat{\pi}_2 - (\pi_1 - \pi_2))/\hat{\sigma} \xrightarrow{d} N(0,1)$, where $\hat{\sigma}$ replaces π_i by $\hat{\pi}_i$; so $(\hat{\pi}_1 - \hat{\pi}_2) \pm z_{\alpha/2}\hat{\sigma}$ is a Wald interval.

Proof.

Step 1. Linearity and independence give the mean and variance, using $\text{Var}(\hat{\pi}_i) = \pi_i(1-\pi_i)/n_i$ (Proposition 1.8).

Step 2. [Supplement] The two standardized proportions are independent and asymptotically $N(0,1)$ (CLT), so is the standardized difference; $\hat{\sigma}/\sigma \xrightarrow{p} 1$ and Slutsky completes the proof. ■

As with (1.11), coverage is usually below nominal near 0 or 1.

3.1.4 Interval estimation of relative risk

The sample relative risk $r = \hat{\pi}_1/\hat{\pi}_2$ is also better handled on the log scale, with

$$\sigma(\log r) = \left[\frac{1-\pi_1}{\pi_1 n_1} + \frac{1-\pi_2}{\pi_2 n_2}\right]^{1/2} \tag{3.3}$$

(Corollary 3.5(b)); exponentiate $\log r \pm z_{\alpha/2}\hat{\sigma}(\log r)$.

Example 3.3 (Swedish aspirin study: relative risk and difference)

Proportions 0.0409 (placebo), 0.0266 (aspirin).
(a) $r = 1.54$, interval (0.86, 2.75).
(b) Difference 0.014, interval $(-0.005, 0.033)$.
No effect is plausible; the larger study of Example 2.7 shows a benefit.

3.1.5 Deriving standard errors with the delta method*

Theorem 3.4 (Delta method)

Suppose $\sqrt{n}(T_n - \theta) \xrightarrow{d} N(0, \sigma^2)$ and g is differentiable at θ. Then

$$\sqrt{n}\,[g(T_n) - g(\theta)] \xrightarrow{d} N\big(0,\, [g'(\theta)]^2\sigma^2\big).$$

If moreover g' is continuous at θ, $g'(\theta) \neq 0$, and $\hat{\sigma} \xrightarrow{p} \sigma > 0$, then $\sqrt{n}[g(T_n) - g(\theta)]/(|g'(T_n)|\hat{\sigma}) \xrightarrow{d} N(0,1)$.

Proof. [Supplement]

Step 1. Let $h(t) = \frac{g(t)-g(\theta)}{t-\theta} - g'(\theta)$ for $t \neq \theta$, $h(\theta) = 0$; h is continuous at θ. Since $T_n \xrightarrow{p} \theta$, $h(T_n) \xrightarrow{p} 0$.

Step 2. $\sqrt{n}[g(T_n) - g(\theta)] = [g'(\theta) + h(T_n)]\sqrt{n}(T_n - \theta) \xrightarrow{d} g'(\theta)N(0,\sigma^2)$ by Slutsky.

Step 3. $|g'(T_n)|\hat{\sigma} \xrightarrow{p} |g'(\theta)|\sigma > 0$, and Slutsky again gives the standardized version. ∎

Locally g is linear with slope $g'(\theta)$, so the spread of $g(T_n)$ is $|g'(\theta)|$ times that of T_n (Figure 1).

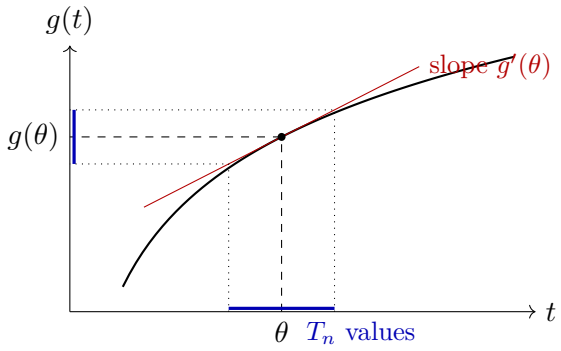


Figure 1: The delta method: near θ, g is almost linear, so a band of T_n values maps to a band of $g(T_n)$ values about $|g'(\theta)|$ times as wide.

3.1.6 Delta method applied to sample logit*

Corollary 3.5 (Sample logit and log proportion)

Let $\hat{\pi} = y/n$ with $y \sim \text{bin}(n, \pi)$, $0 < \pi < 1$.
(a) The sample logit satisfies

$$\sqrt{n}\Big[\log\frac{\hat{\pi}}{1-\hat{\pi}} - \log\frac{\pi}{1-\pi}\Big] \xrightarrow{d} N\Big(0, \frac{1}{\pi(1-\pi)}\Big).$$

(b) $\sqrt{n}(\log\hat{\pi} - \log\pi) \xrightarrow{d} N(0, (1-\pi)/\pi)$; hence for independent samples the log relative risk has asymptotic variance (3.3).

Proof.

Step (a). $\sqrt{n}(\hat{\pi} - \pi) \xrightarrow{d} N(0, \pi(1-\pi))$, and $g(\pi) = \log[\pi/(1-\pi)]$ has $g'(\pi) = 1/[\pi(1-\pi)]$; Theorem 3.4 gives variance $\pi(1-\pi)[\pi(1-\pi)]^{-2}$.

Step (b). With $g = \log$, $g'(\pi) = 1/\pi$ and the variance is $(1-\pi)/\pi$. Dividing by n_i and adding over the two independent samples gives (3.3). ■

The asymptotic variance is that of the limiting normal; the true variance does not exist, since $\hat{\pi} \in \{0, 1\}$ with positive probability.

3.1.7 Delta method for log odds ratio*

Theorem 3.6 (Multivariate delta method for multinomial proportions)

Let $(n_1, \ldots, n_c)$ be multinomial$(n, \boldsymbol{\pi})$, $\hat{\boldsymbol{\pi}} = \mathbf{n}/n$, and let g be differentiable at $\boldsymbol{\pi}$ with $\phi_i = \partial g/\partial\pi_i$. Define

$$\sigma^2 = \sum_i \pi_i\phi_i^2 - \Big(\sum_i \pi_i\phi_i\Big)^2. \tag{3.4}$$

If $\sigma > 0$ then $\sqrt{n}[g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi})]/\sigma \xrightarrow{d} N(0, 1)$, and the same holds with σ replaced by its ML estimate $\hat{\sigma}$ (substituting $\hat{\pi}_i$ for π_i in (3.4)).

Proof. [Supplement]

Step 1. By the multivariate CLT, $\sqrt{n}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}) \xrightarrow{d} \mathbf{z} \sim N_c(\mathbf{0}, \Sigma)$, $\Sigma = \text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top$ (Proposition 1.2).

Step 2. Differentiability gives $\sqrt{n}[g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi})] = \boldsymbol{\phi}^\top\sqrt{n}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}) + o_p(1) \xrightarrow{d} \boldsymbol{\phi}^\top\mathbf{z}$, since $\|\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}\| = O_p(n^{-1/2})$.

Step 3. $\boldsymbol{\phi}^\top\mathbf{z} \sim N(0, \boldsymbol{\phi}^\top\Sigma\boldsymbol{\phi})$, and $\boldsymbol{\phi}^\top\Sigma\boldsymbol{\phi} = \sum_i \pi_i\phi_i^2 - (\sum_i \pi_i\phi_i)^2 = \sigma^2$.

Step 4. If the ϕ_i are continuous, $\hat{\sigma} \xrightarrow{p} \sigma$ (WLLN), and Slutsky applied to the product of the standardized statistic and $\sigma/\hat{\sigma} \xrightarrow{p} 1$ gives the limit $N(0, 1)$. ■

Corollary 3.7 (Standard error of the log odds ratio)

For a multinomial 2×2 table, $\log\hat{\theta}$ has asymptotic standard error $\big(\sum_i\sum_j 1/(n\pi_{ij})\big)^{1/2}$, estimated by (3.1).

Proof.

Step 1. $g(\boldsymbol{\pi}) = \log\pi_{11} + \log\pi_{22} - \log\pi_{12} - \log\pi_{21}$ has $\phi_{ij} = \pm 1/\pi_{ij}$ (+ on the diagonal).

Step 2. $\sum\sum \pi_{ij}\phi_{ij} = 1-1-1+1 = 0$ and $\sum\sum \pi_{ij}\phi_{ij}^2 = \sum\sum 1/\pi_{ij}$; Theorem 3.6 gives $\sigma/\sqrt{n} = (\sum\sum 1/(n\pi_{ij}))^{1/2}$, estimated by $(\sum\sum 1/n_{ij})^{1/2}$. ■

The same holds for independent binomial rows (a difference of two sample logits, Corollary 3.5(a)). A Wald interval for θ itself is not recommended.

3.2 Testing independence in two-way contingency tables

Motivation. Independence, $\pi_{ij} = \pi_{i+}\pi_{+j}$, is the starting point of contingency-table analysis, tested by X^2 and G^2.

Under independent multinomial rows the hypothesis is homogeneity; the same tests apply.

3.2.1 Pearson and likelihood-ratio chi-squared tests

Proposition 3.8 (ML estimates under independence)

Under H_0, the ML estimates are $\hat\pi_{i+} = n_{i+}/n$, $\hat\pi_{+j} = n_{+j}/n$, so the estimated expected frequencies are

$$\hat\mu_{ij} = n\hat\pi_{i+}\hat\pi_{+j} = \frac{n_{i+}n_{+j}}{n}.$$

Proof. The kernel $\prod_i\prod_j(\pi_{i+}\pi_{+j})^{n_{ij}} = \prod_i \pi_{i+}^{n_{i+}} \cdot \prod_j \pi_{+j}^{n_{+j}}$ is a product of two multinomial kernels in separate parameters, each maximized by sample proportions (Proposition 1.14). ■

$$X^2 = \sum_i\sum_j \frac{(n_{ij}-\hat\mu_{ij})^2}{\hat\mu_{ij}}, \qquad G^2 = 2\sum_i\sum_j n_{ij}\log\frac{n_{ij}}{\hat\mu_{ij}}. \tag{3.5}$$

Theorem 3.9 (Null distribution of X^2 and G^2 for independence)

Under H_0 with all $\pi_{i+}, \pi_{+j} > 0$, X^2 and G^2 both converge in distribution to $\chi^2_{(I-1)(J-1)}$, and $X^2 - G^2 \xrightarrow{p} 0$.

Proof. [Supplement] Apply Theorem 1.18 to the $c = IJ$ cells with the $p = (I-1)+(J-1)$ parameters $\pi_{1+},\ldots,\pi_{I-1,+},\pi_{+1},\ldots,\pi_{+,J-1}$.

Step 1 (rank). With $a_i = \pi_{i+}$, $b_j = \pi_{+j}$, a tangent direction is $d\pi_{ij} = b_j\,da_i + a_i\,db_j$ with $\sum_i da_i = \sum_j db_j = 0$. If all $d\pi_{ij} = 0$, summing over j gives $da_i = 0$, and then $db_j = 0$; so the Jacobian has rank p.

Step 2 (df). Hence $X^2 \xrightarrow{d} \chi^2_{IJ-1-p}$ with $IJ-1-(I-1)-(J-1) = (I-1)(J-1)$, and $G^2 - X^2 \xrightarrow{p} 0$. ■

Fisher (1922) corrected Pearson's df $= IJ-1$. The limits also hold for Poisson and product multinomial sampling; X^2 converges faster than G^2 (small-sample methods: Section 3.5).

3.2.2 Education and religious fundamentalism example

Example 3.10 (Education and religious beliefs)

1996 GSS: count, $\hat{\mu}_{ij}$, and standardized residual (3.6).

Highest degree	Fundamentalist		Moderate		Liberal		Total
< High school	178 (137.8)	4.5	138 (161.5)	−2.6	108 (124.7)	−1.9	424
HS / junior college	570 (539.5)	2.6	648 (632.1)	1.3	442 (488.4)	−4.0	1660
Bachelor / graduate	138 (208.7)	−6.8	252 (244.5)	0.7	252 (188.9)	6.3	642
Total	886		1038		802		2726

$X^2 = 69.2$, $G^2 = 69.8$, df $= 4$, $P < 0.0001$.

3.3 Following-up chi-squared tests

Motivation. Residuals, partitions of G^2 and odds ratios reveal the nature of an association that a test only detects.

3.3.1 Pearson and standardized residuals

Definition (Pearson and standardized residuals)

The *Pearson residual* and *standardized Pearson residual* for independence are

$$e_{ij} = \frac{n_{ij} - \hat{\mu}_{ij}}{\hat{\mu}_{ij}^{1/2}}, \qquad r_{ij} = \frac{n_{ij} - \hat{\mu}_{ij}}{[\hat{\mu}_{ij}(1 - p_{i+})(1 - p_{+j})]^{1/2}}, \tag{3.6}$$

with $p_{i+} = n_{i+}/n$, $p_{+j} = n_{+j}/n$. Note $\sum_i \sum_j e_{ij}^2 = X^2$.

Theorem 3.11 (Asymptotic variance of Pearson residuals)

Under independence, $e_{ij} \xrightarrow{d} N(0, (1 - \pi_{i+})(1 - \pi_{+j}))$; hence $r_{ij} \xrightarrow{d} N(0, 1)$. The average asymptotic variance of the e_{ij} is $(I - 1)(J - 1)/(IJ) < 1$.

Proof. [Supplement] Use the notation of the proof of Theorem 1.18 with cells (i, j), $a_i = \pi_{i+}$, $b_j = \pi_{+j}$, and unit vectors $\mathbf{r} = (\sqrt{a_i})_i$, $\mathbf{s}_c = (\sqrt{b_j})_j$, so $\mathbf{s} = \mathbf{r} \otimes \mathbf{s}_c$.

Step 1. By that proof the Pearson residuals are $(I - H)\mathbf{z}_n + o_p(1)$ with limiting covariance $I - H - \mathbf{s}\mathbf{s}^\top$, H the projection onto the column space of A.

Step 2. A tangent direction $d\pi_{ij} = b_j\, da_i + a_i\, db_j$ gives $D^{-1/2} d\boldsymbol{\pi}$ with entries $\sqrt{b_j}\, x_i + \sqrt{a_i}\, y_j$, $x_i = da_i/\sqrt{a_i}$, $y_j = db_j/\sqrt{b_j}$, where $\mathbf{x} \perp \mathbf{r}$, $\mathbf{y} \perp \mathbf{s}_c$. So the column space is $(\mathbf{r}^\perp \otimes \mathbf{s}_c) \oplus (\mathbf{r} \otimes \mathbf{s}_c^\perp)$, two orthogonal subspaces, and

$$H = (I_I - \mathbf{r}\mathbf{r}^\top) \otimes \mathbf{s}_c\mathbf{s}_c^\top + \mathbf{r}\mathbf{r}^\top \otimes (I_J - \mathbf{s}_c\mathbf{s}_c^\top).$$

Step 3. The (ij, ij) entries of H and $\mathbf{s}\mathbf{s}^\top$ are $(1 - a_i)b_j + a_i(1 - b_j)$ and $a_i b_j$, so e_{ij} has asymptotic variance $1 - a_i - b_j + a_i b_j = (1 - a_i)(1 - b_j)$.

Step 4. Slutsky gives $r_{ij} \xrightarrow{d} N(0, 1)$, and $\frac{1}{IJ} \sum_i \sum_j (1 - a_i)(1 - b_j) = \frac{(I-1)(J-1)}{IJ}$. ■

Comparing e_{ij} with $N(0, 1)$ is conservative; $|r_{ij}|$ above about 2 or 3 indicates lack of fit in that cell.

3.3.2 Education and religious fundamentalism revisited

Example 3.12 (Residuals for education and religion)

$r_{11} = (178 - 137.8)/[137.8(1 - 0.156)(1 - 0.325)]^{1/2} = 4.5$. Large positive residuals at (less than HS, fundamentalist) and (bachelor+, liberal), large negative at the other two corners; the corner 2×2 odds ratio is $\frac{178 \times 252}{108 \times 138} = 3.0$.

3.3.3 Partitioning chi-squared

Since independent $\chi^2_{\nu_1}$ and $\chi^2_{\nu_2}$ sum to $\chi^2_{\nu_1+\nu_2}$, a statistic can be split into components showing which categories drive an association.

Theorem 3.13 (Exact partition of G^2 by successive columns)

For an $I \times J$ table, for $j = 1, \ldots, J-1$ let G^2_j be the G^2 statistic for independence in the $I \times 2$ table whose first column merges columns $1, \ldots, j$ and whose second column is column $j+1$. Then

$$G^2 = \sum_{j=1}^{J-1} G^2_j,$$

and each G^2_j has $\mathrm{df} = I - 1$, so the df also add up to $(I-1)(J-1)$.

Proof. [Supplement]

Step 1. With $\Phi(x) = x \log x$, for any table with counts m_{ij}, margins m_{i+}, m_{+j} and total m, (3.5) gives

$$G^2/2 = \sum_i \sum_j \Phi(m_{ij}) - \sum_i \Phi(m_{i+}) - \sum_j \Phi(m_{+j}) + \Phi(m),$$

since $\log(m_{ij}/\hat{\mu}_{ij}) = \log m_{ij} + \log m - \log m_{i+} - \log m_{+j}$.

Step 2. Let $S_{ij} = \sum_{l \le j} n_{il}$ and $S_{+j} = \sum_i S_{ij}$. The jth subtable has cells S_{ij}, $n_{i,j+1}$, row totals $S_{i,j+1}$, column totals S_{+j}, $n_{+,j+1}$ and total $S_{+,j+1}$, so $G^2_j/2$ equals

$$\sum_i [\Phi(S_{ij}) + \Phi(n_{i,j+1}) - \Phi(S_{i,j+1})] - \Phi(S_{+j}) - \Phi(n_{+,j+1}) + \Phi(S_{+,j+1}).$$

Step 3. Summing over j, $\Phi(S_{ij}) - \Phi(S_{i,j+1})$ telescopes to $\Phi(n_{i1}) - \Phi(n_{i+})$, which with $\sum_{j \ge 2} \Phi(n_{ij})$ gives $\sum_j \Phi(n_{ij}) - \Phi(n_{i+})$; likewise the column terms give $\Phi(n) - \sum_j \Phi(n_{+j})$. The total is $G^2/2$ by Step 1. ■

Repeating this on rows gives Lancaster's partition into $(I-1)(J-1)$ components with $\mathrm{df} = 1$, from the 2×2 tables

$$\begin{matrix} \sum_{a<i} \sum_{b<j} n_{ab} & \sum_{a<i} n_{aj} \\ \sum_{b<j} n_{ib} & n_{ij} \end{matrix} \qquad i = 2, \ldots, I, \; j = 2, \ldots, J. \tag{3.7}$$

Comparing each column with the last does *not* give an additive partition (Section 9.2.4 views partitions as nested models).

3.3.4 Origin of schizophrenia example

Example 3.14 (School of psychiatric thought and origin of schizophrenia)

Schools (eclectic, medical, psychoanalytic) by ascribed origin (biogenic, environmental, combination): $G^2 = 23.04$, df $= 4$. By (3.7), Ecl vs. Med components are 0.29 and 1.36; (Ecl+Med) vs. Psy components are 12.95 (Bio vs. Env) and 8.43 ((Bio+Env) vs. Com), summing to 23.04. Psychoanalysts more often ascribe a combined or environmental origin.

3.3.5 Rules for partitioning

Necessary conditions for independent components (Goodman; Lancaster):

(a) the subtable df sum to the df of the full table;

(b) each cell count of the full table is a cell count in exactly one subtable;

(c) each marginal total of the full table is a marginal total of exactly one subtable.

Exact partitions hold for G^2 but not X^2.

3.3.6 Limitations of chi-squared tests

They only measure evidence, need large samples, and are invariant to reordering categories, so they ignore ordinal information (Section 3.4).

3.3.7 Why consider independence?

If independence approximately holds, $p_{i+}p_{+j}$ can beat p_{ij} in MSE: biased, but with fewer parameters.

Proposition 3.15 (MSE of the two estimators)

For multinomial sampling,

$$\sum_i \sum_j E(p_{ij} - \pi_{ij})^2 = \frac{1 - \sum_i \sum_j \pi_{ij}^2}{n}, \qquad \sum_i \sum_j E(p_{i+}p_{+j} - \pi_{ij})^2 \to \sum_i \sum_j (\pi_{i+}\pi_{+j} - \pi_{ij})^2.$$

Under independence, $n \sum \sum E(p_{i+}p_{+j} - \pi_{ij})^2 \to \sum_i \sum_j [\pi_{+j}^2 \pi_{i+}(1 - \pi_{i+}) + \pi_{i+}^2 \pi_{+j}(1 - \pi_{+j})]$.

Proof. [Supplement]

Step 1. p_{ij} is unbiased with $\mathrm{Var}(p_{ij}) = \pi_{ij}(1 - \pi_{ij})/n$; sum over cells.

Step 2. With $a = p_{i+} - \pi_{i+}$, $b = p_{+j} - \pi_{+j}$, $p_{i+}p_{+j} - \pi_{ij} = (\pi_{i+}\pi_{+j} - \pi_{ij}) + \pi_{+j}a + \pi_{i+}b + ab$. Since $E(a^2), E(b^2) = O(n^{-1})$ and $E(a^2b^2) \le [E(a^4)E(b^4)]^{1/2} = O(n^{-2})$, all but the first term vanish in mean square.

Step 3. Under independence the first term is 0, and $nE(\pi_{+j}a + \pi_{i+}b)^2 = \pi_{+j}^2 \pi_{i+}(1-\pi_{i+}) + \pi_{i+}^2 \pi_{+j}(1-\pi_{+j}) + 2\pi_{i+}\pi_{+j}\, n\,\mathrm{Cov}(a, b)$, where $n\,\mathrm{Cov}(p_{i+}, p_{+j}) = \pi_{ij} - \pi_{i+}\pi_{+j} = 0$; the terms involving ab are $o(n^{-1})$ by Cauchy–Schwarz. ∎

Example 3.16 (A 3×3 illustration)

Let $\pi_{i+} = \pi_{+j} = 1/3$ and $\pi_{ij} = \frac{1}{9}[1 + \delta(i-2)(j-2)]$.
(a) At $\delta = 0$ the total MSE is $8/(9n)$ for p_{ij} and $\approx 4/(9n)$ for $p_{i+}p_{+j}$.

(b) For $\delta \neq 0$ the latter tends to $4\delta^2/81$, yet still wins for moderate n (e.g. $n = 500$, $\delta = 0.1$).

3.4 Two-way tables with ordered classifications

Motivation. For ordinal variables, one-parameter trend statistics give df $= 1$ tests with much more power against monotone alternatives.

3.4.1 Linear trend alternative to independence

With row scores $u_1 \leq \cdots \leq u_I$ and column scores $v_1 \leq \cdots \leq v_J$, let r be the sample correlation of the scores over the n subjects:

$$r = \frac{\sum\sum u_i v_j n_{ij} - (\sum_i u_i n_{i+})(\sum_j v_j n_{+j})/n}{\left[\sum_i u_i^2 n_{i+} - \frac{(\sum_i u_i n_{i+})^2}{n}\right]^{1/2}\left[\sum_j v_j^2 n_{+j} - \frac{(\sum_j v_j n_{+j})^2}{n}\right]^{1/2}}.$$

Definition (Linear trend statistic)

$M^2 = (n-1)r^2$ (Mantel, 1963), referred to χ^2_1; for one-sided alternatives use $M = \sqrt{n-1}\, r$ with the normal tail.

Theorem 3.17 (Null distribution of M^2)

Under independence of X and Y (multinomial sampling, each score variable nondegenerate), $\sqrt{n}\, r \xrightarrow{d} N(0,1)$; hence $M \xrightarrow{d} N(0,1)$ and $M^2 \xrightarrow{d} \chi^2_1$.

Proof. [Supplement] Let (U_k, V_k), $k \leq n$, be the subjects' score pairs: i.i.d., U independent of V, with means μ, ν and variances $\sigma^2, \tau^2 > 0$.

Step 1. $S_{UV} = \frac{1}{n}\sum_k W_k - (\bar{U} - \mu)(\bar{V} - \nu)$ with $W_k = (U_k - \mu)(V_k - \nu)$ i.i.d., $E(W_k) = 0$ and $\mathrm{Var}(W_k) = \sigma^2\tau^2$ by independence; so $n^{-1/2}\sum_k W_k \xrightarrow{d} N(0, \sigma^2\tau^2)$.

Step 2. $\sqrt{n}(\bar{U} - \mu)(\bar{V} - \nu) = O_p(1)o_p(1) \xrightarrow{p} 0$, so $\sqrt{n}S_{UV} \xrightarrow{d} N(0, \sigma^2\tau^2)$.

Step 3. $S_{UU} \xrightarrow{p} \sigma^2$, $S_{VV} \xrightarrow{p} \tau^2$, so by Slutsky $\sqrt{n}\, r \xrightarrow{d} N(0,1)$; as $\sqrt{(n-1)/n} \to 1$, $M \xrightarrow{d} N(0,1)$ and $M^2 \xrightarrow{d} \chi^2_1$. ■

3.4.2 Job satisfaction example revisited

Example 3.18 (Linear trend test for income and job satisfaction)

For Example 2.17 ($n = 96$), $X^2 = 6.0$, $G^2 = 6.8$, df $= 9$ ($P \approx 0.7$). With scores $1, 2, 3, 4$ and income midpoints $7.5, 20, 32.5, 60$, $r = 0.200$, $M^2 = 3.81$ ($P = 0.051$) and $M = 1.95$ (one-sided $P = 0.026$): much stronger evidence.

3.4.3 Monotone trend alternatives to independence

Ordinal measures such as gamma (Section 2.4.4) give $z = \hat{\gamma}/\mathrm{SE}$; for Example 2.17, $\hat{\gamma} = 0.221$, SE $= 0.117$, $z = 1.89$, 95% interval $(-0.01, 0.45)$.

3.4.4 Extra power with ordinal tests

X^2 and G^2 spread power over $(I-1)(J-1)$ parameters; M^2 and $(\hat{\gamma}/\mathrm{SE})^2$ concentrate it on one, so a comparable value lies much farther in the χ^2_1 tail (Section 6.4).

3.4.5 Choice of scores

Any scores chosen before seeing the data give a valid test, but poor scores give an insensitive one; linear transformations leave M^2 unchanged.

Example 3.19 (Maternal drinking and congenital malformations)

Malformation (present/absent) by drinks per day (0, <1, 1–2, 3–5, ≥ 6), with present counts $48, 38, 5, 1, 1$ among 17,114, 14,502, 793, 127, 38 subjects.
(a) Scores $0, 0.5, 1.5, 4.0, 7.0$: $M^2 = 6.57$, $P = 0.010$.
(b) Scores $1, 2, 3, 4, 5$: $M^2 = 1.83$, $P = 0.18$.
(c) Midranks: $M^2 = 0.35$, $P = 0.55$; they treat the sparse last three categories as nearly equal, which is inappropriate.

Use scores reflecting distances and check sensitivity. With midranks for both variables, r is *Spearman's rho*.

3.4.6 Trend tests for $I \times 2$ and $2 \times J$ tables

For $2 \times J$ with $u = (0, 1)$, M^2 compares the two row means of the Y scores; with midranks it is the *Wilcoxon–Mann–Whitney* test. For $I \times 2$ it is the *Cochran–Armitage trend test* (Section 5.3.5).

3.4.7 Nominal–ordinal tables

With a nominal variable of more than two categories, compare mean scores of the ordinal variable across categories (Section 7.5.3).

3.5 Small-sample tests of independence

Motivation. Conditioning on the margins eliminates nuisance parameters and gives the exact hypergeometric null distribution.

3.5.1 Fisher's exact test for 2×2 tables

Theorem 3.20 (Hypergeometric null distribution)

Under independence, for Poisson, multinomial, or independent binomial sampling, conditional on both sets of marginal totals,

$$p(t) = P(n_{11} = t) = \frac{\binom{n_{1+}}{t}\binom{n_{2+}}{n_{+1}-t}}{\binom{n}{n_{+1}}}, \qquad m_- \leq t \leq m_+, \tag{3.8}$$

where $m_- = \max(0,\, n_{1+} + n_{+1} - n)$ and $m_+ = \min(n_{1+},\, n_{+1})$.

Proof.

Step 1. By Theorem 2.3, given row totals, $n_{11} \sim \text{bin}(n_{1+}, \pi)$ and $n_{21} \sim \text{bin}(n_{2+}, \pi)$ independently under H_0, and $n_{+1} \sim \text{bin}(n, \pi)$.

Step 2. So $P(n_{11} = t \mid n_{+1})$ is $\binom{n_{1+}}{t}\binom{n_{2+}}{n_{+1}-t}\pi^{n_{+1}}(1-\pi)^{n-n_{+1}}$ divided by $\binom{n}{n_{+1}}\pi^{n_{+1}}(1-\pi)^{n-n_{+1}}$; the powers cancel, leaving (3.8).

Step 3. $0 \le t \le n_{1+}$ and $0 \le n_{+1} - t \le n_{2+}$ give $m_- \le t \le m_+$. ■

For $H_a\colon \theta > 1$, $P = P(n_{11} \ge t_o)$.

Proposition 3.21 (Orderings of 2×2 tables with fixed margins)

With all margins fixed, the sample odds ratio and the difference of proportions $n_{11}/n_{1+} - n_{21}/n_{2+}$ are increasing in n_{11}, and

$$X^2 = \frac{n^3\big(n_{11} - E(n_{11})\big)^2}{n_{1+}n_{2+}n_{+1}n_{+2}}, \qquad E(n_{11}) = \frac{n_{1+}n_{+1}}{n}.$$

Hence one-sided P-values agree for all three orderings, and $P(X^2 \ge X_o^2) = P(|n_{11} - E(n_{11})| \ge |t_o - E(n_{11})|)$.

Proof. [Supplement]

Step 1. With $n_{12} = n_{1+} - n_{11}$, $n_{21} = n_{+1} - n_{11}$, $n_{22} = n - n_{1+} - n_{+1} + n_{11}$, the numerator of $\hat\theta$ increases and its denominator decreases in n_{11}, and $n_{11}/n_{1+} - (n_{+1} - n_{11})/n_{2+}$ increases.

Step 2. $n_{11}n_{22} - n_{12}n_{21} = n\,n_{11} - n_{1+}n_{+1} = n[n_{11} - E(n_{11})]$ by expanding.

Step 3. In (3.5) all four $n_{ij} - \hat\mu_{ij}$ equal $\pm(n_{11}n_{22} - n_{12}n_{21})/n$ and $\sum_{i,j} 1/\hat\mu_{ij} = n^3/(n_{1+}n_{2+}n_{+1}n_{+2})$, so $X^2 = n(n_{11}n_{22} - n_{12}n_{21})^2/(n_{1+}n_{2+}n_{+1}n_{+2})$; Step 2 gives the display. ■

3.5.2 Fisher's tea drinker

Example 3.22 (Fisher's tea-tasting experiment)

A lady tasted 8 cups, 4 milk-first, and named 4 as milk-first, getting 3 right ($n_{11} = 3$, all margins 4, fixed by design). For $H_a\colon \theta > 1$,

$$P(n_{11} = 3) = \frac{\binom{4}{3}\binom{4}{1}}{\binom{8}{4}} = \frac{16}{70} = 0.229, \qquad P(n_{11} = 4) = \frac{1}{70} = 0.014,$$

so $P = 0.243$: too small an experiment to establish her ability.

3.5.3 Two-sided P-values for Fisher's exact test

Two-sided criteria include:

(a) sum $p(t)$ over t with $p(t) \le p(t_o)$ (the most popular);

(b) $P(|n_{11} - E(n_{11})| \ge |t_o - E(n_{11})|)$, which equals $P(X^2 \ge X_o^2)$ by Proposition 3.21;

(c) $2\min[P(n_{11} \ge t_o), P(n_{11} \le t_o)]$, which can exceed 1;

(d) the smaller tail plus the largest attainable probability in the other tail not exceeding it (Blaker).

3.5.4 Discreteness and conservatism issues

In the tea experiment one-sided P-values can only be 0.014, 0.243, 0.757, 0.986, 1, so rejecting at $P \leq 0.05$ has actual size 0.014: the test is *conservative*.

Example 3.23 (Randomized test of exact size 0.05)

Reject if $n_{11} = 4$, and with probability 0.157 if $n_{11} = 3$: $P(\text{reject}) = 0.014 + 0.157(0.229) = 0.05$.

Randomization is unacceptable in practice; instead report the *mid-P-value* $\frac{1}{2}P(n_{11} = t_o) + P(n_{11} > t_o)$ (0.129 for the tea data), rarely much above nominal size.

3.6 Extensions for multiway tables and nontabulated responses

Motivation. Models extend these tests to conditional independence (Chapter 6) and to continuous explanatory variables.

3.6.1 Categorical data need not be contingency tables

Data files usually have one line per subject; the generalized linear models of the next chapter contain this chapter's methods as special cases.